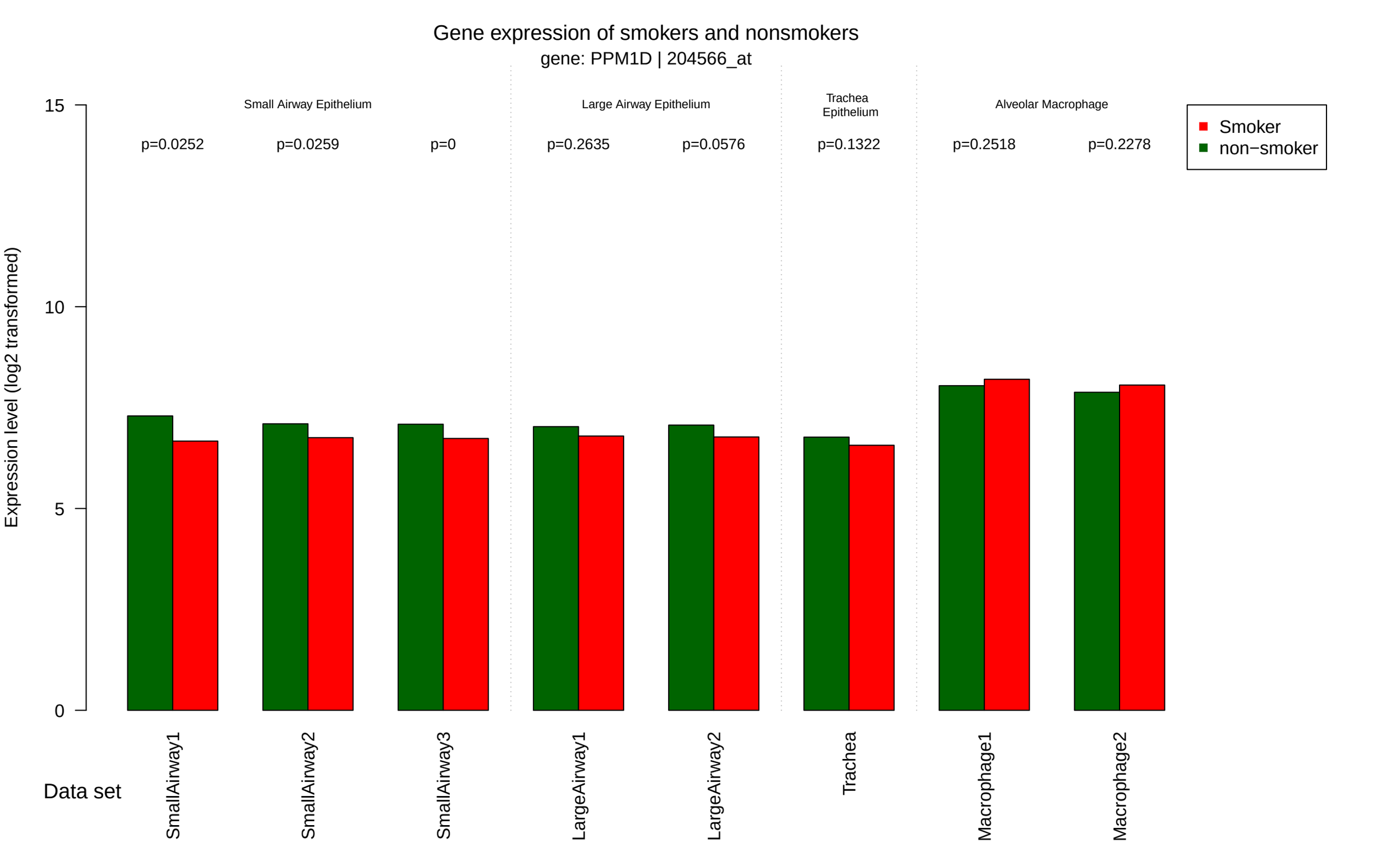
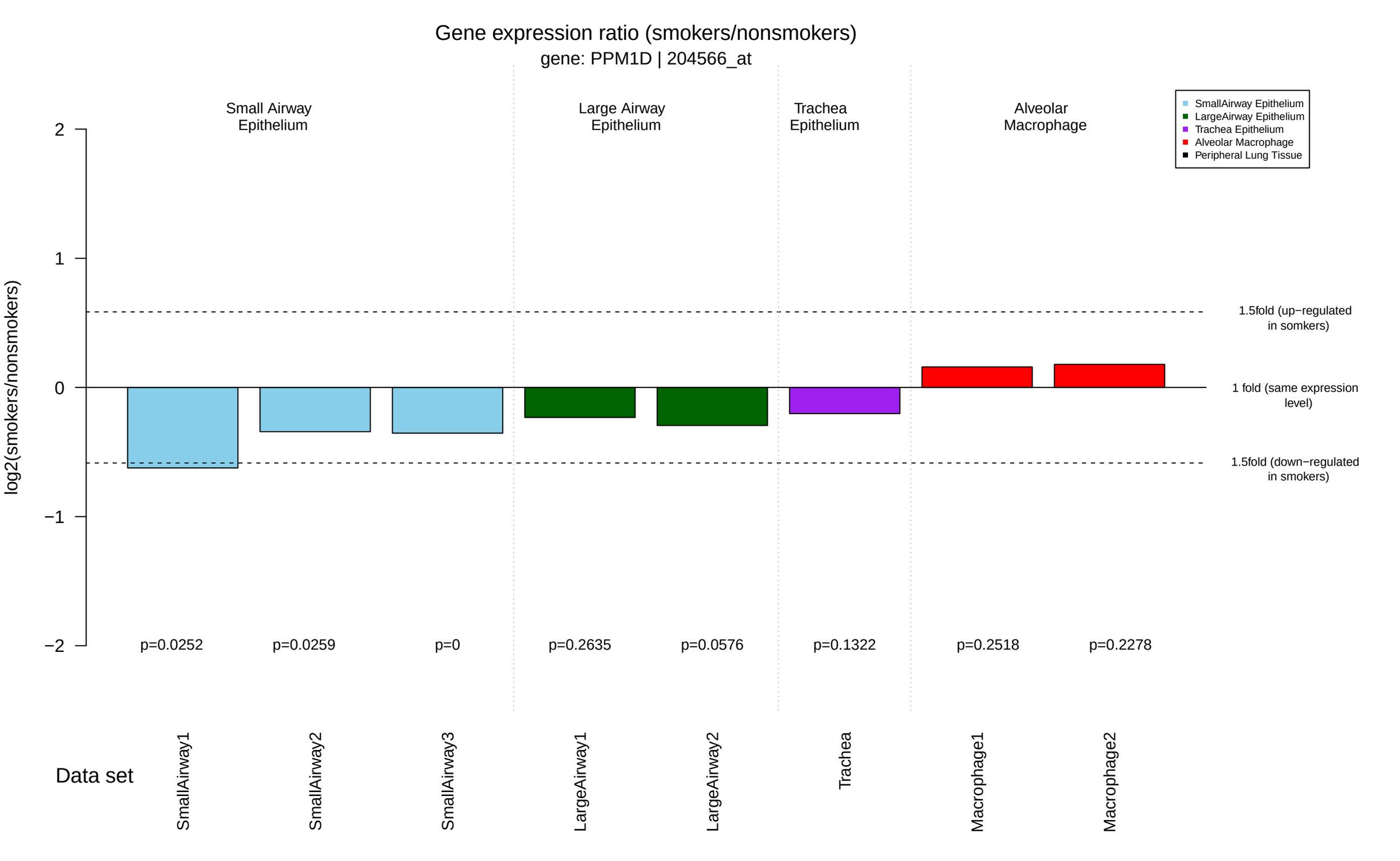
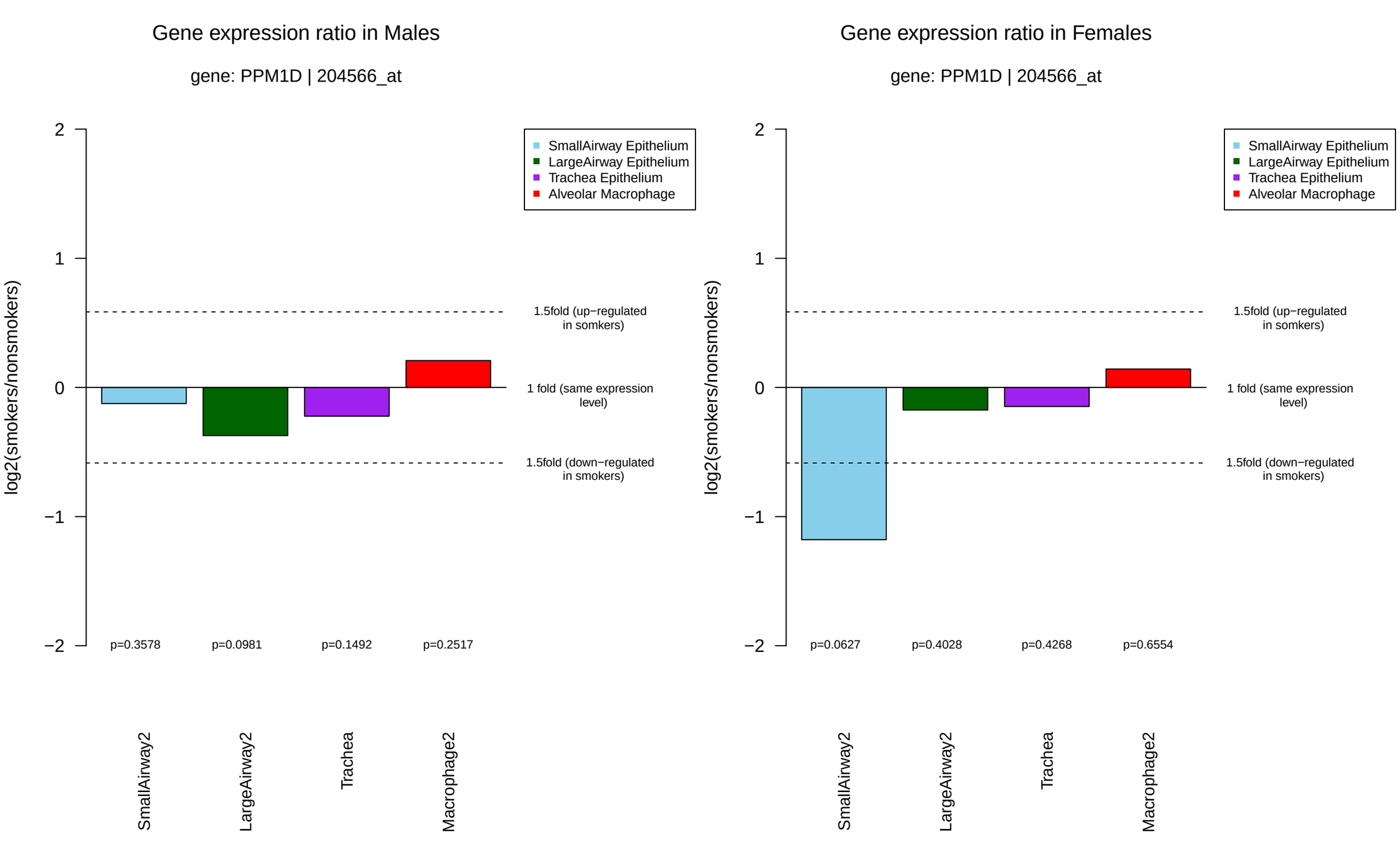
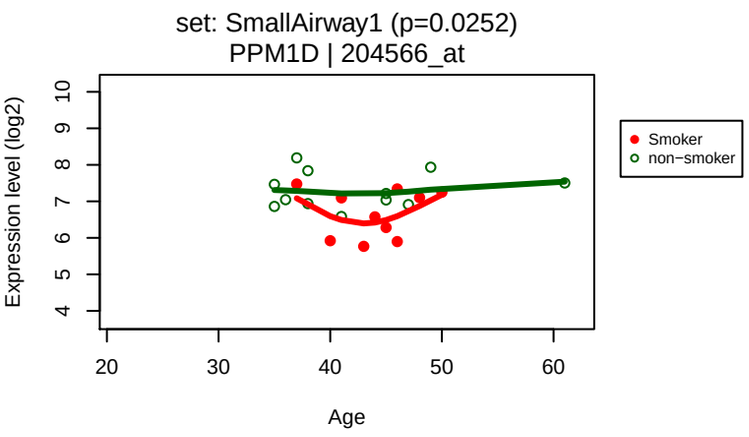
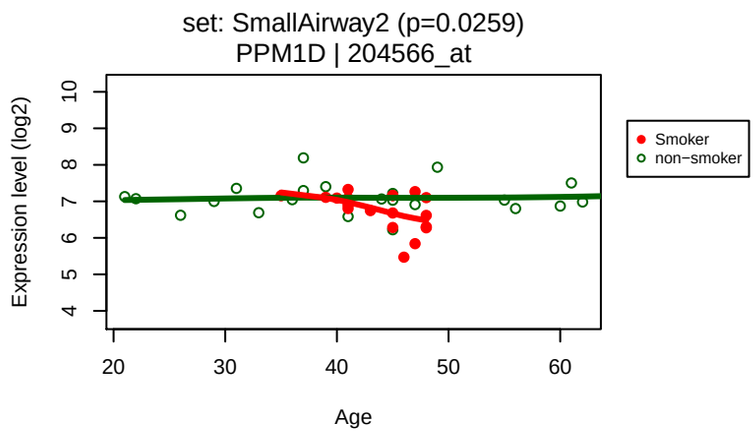
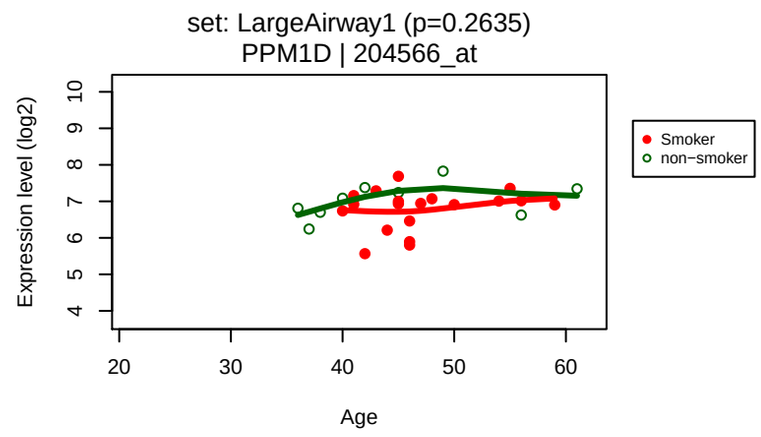

Gene expression of smokers and nonsmokers
gene: PPM1D | 204566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0252
p=0.0259
p=0
p=0.2635
p=0.0576
p=0.1322
p=0.2518
p=0.2278
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

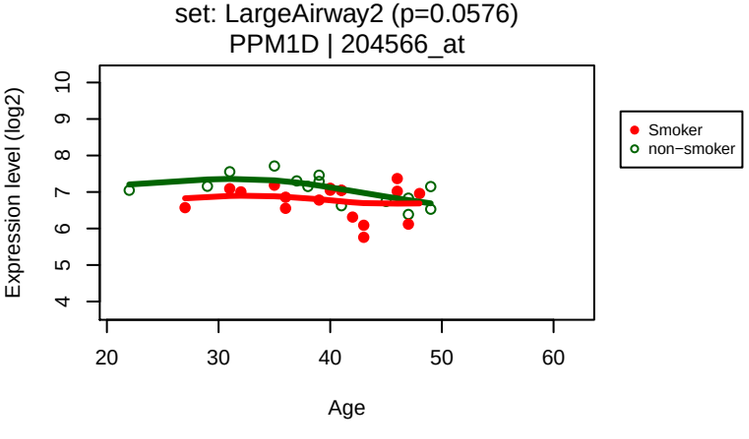

Gene expression ratio (smokers/nonsmokers)
gene: PPM1D | 204566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0252
p=0.0259
p=0
p=0.2635
p=0.0576
p=0.1322
p=0.2518
p=0.2278
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

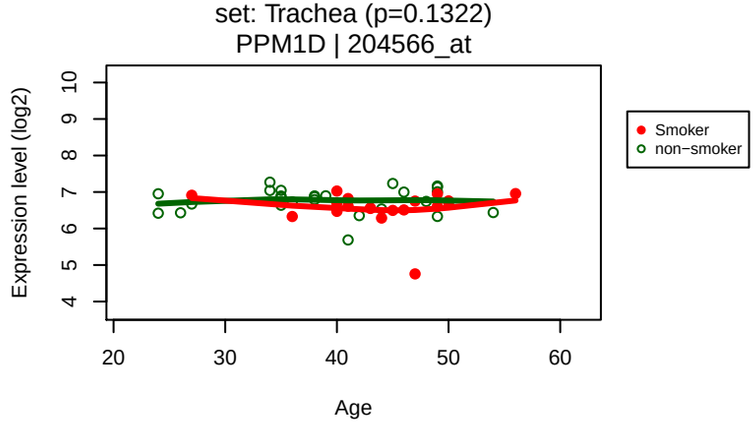

Gene expression ratio in Males
gene: PPM1D | 204566_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3578
p=0.0981
p=0.1492
p=0.2517
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1D | 204566_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0627
p=0.4028
p=0.4268
p=0.6554
SmallAirway2
LargeAirway2
Trachea
Macrophage2

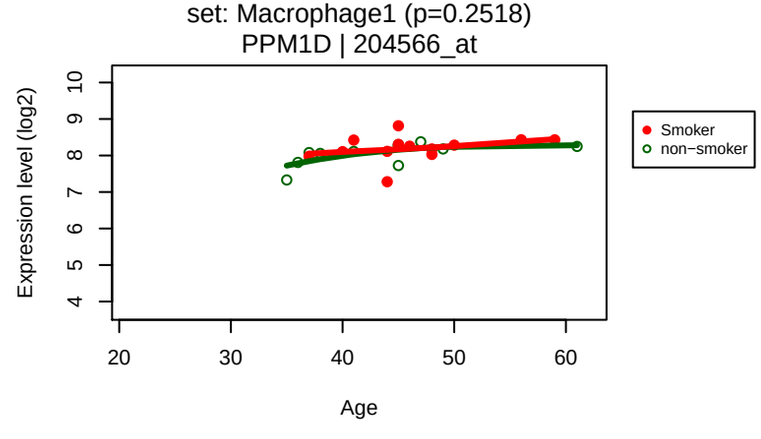

set: SmallAirway1 (p=0.0252)
PPM1D | 204566_at
Expression level (log2)
Age
Smoker
non−smoker

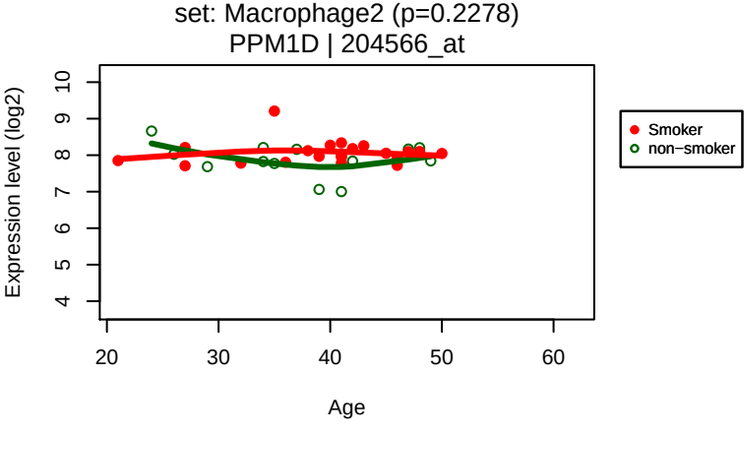

set: SmallAirway2 (p=0.0259)
PPM1D | 204566_at
Expression level (log2)
Age
Smoker
non−smoker

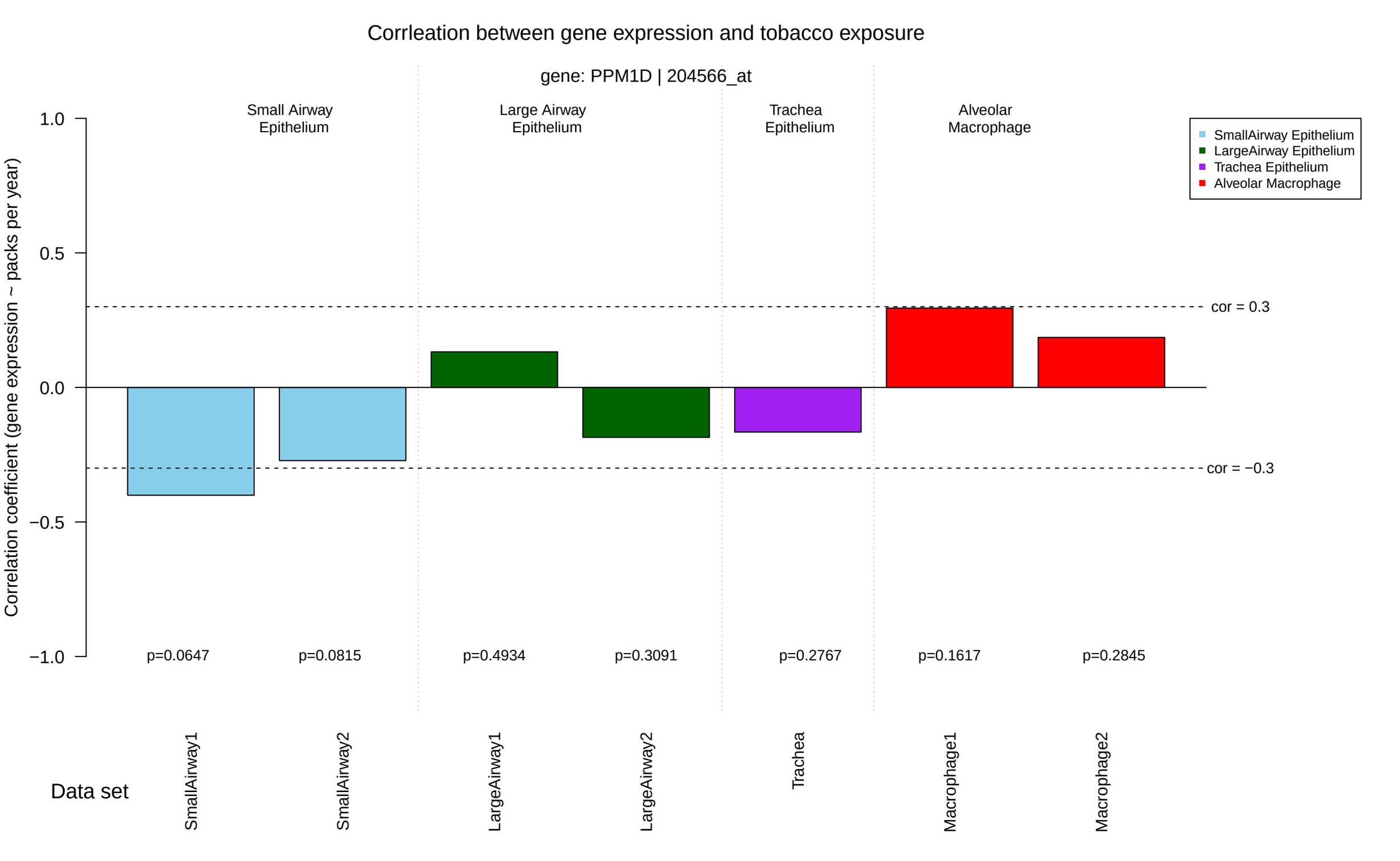

set: LargeAirway1 (p=0.2635)
PPM1D | 204566_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0576)
PPM1D | 204566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1322)
PPM1D | 204566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2518)
PPM1D | 204566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2278)
PPM1D | 204566_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1D | 204566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0647
p=0.0815
p=0.4934
p=0.3091
p=0.2767
p=0.1617
p=0.2845
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2